

Solution Requirements

Date	3 July 2026
Team ID	SWTID-2026-8260
Project Name	A Comprehensive Measure of Well-Being using HDI Prediction System
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Functional Requirements (FR)

1. System should accept user inputs (Life Expectancy, Education, Income)
2. System should preprocess and validate input data
3. System should predict HDI using ML model
4. System should display predicted HDI value
5. System should classify HDI (Low, Medium, High)
6. System should allow repeated predictions

Non-Functional Requirements (NFR)

1. System should respond within 2–3 seconds
2. System should be user-friendly
3. System should ensure data accuracy
4. System should be reliable and consistent
5. System should be accessible via web browser
6. System should handle invalid inputs gracefully

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>Users can access system (optional login)</i>	Medium
2	Authorization	<i>Single user system (admin only)</i>	Low
3	External interfaces	<i>Dataset integration (CSV)</i>	High
4	Transactions processing	<i>Input → Prediction → Output</i>	High
5	Reporting	<i>Display HDI result & category</i>	High
6	Business rules	<i>ML model predicts HDI based on inputs</i>	High
7	Compliance	<i>Data handled securely</i>	Medium
8	<i>Other</i>	<i>Web UI interaction</i>	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	<i>Fast prediction response</i>	<i>< 2 seconds</i>
2	Scalability	<i>Can handle multiple users</i>	<i>Moderate</i>
3	Security & Data Privacy	<i>Basic data protection</i>	<i>Secure input</i>
4	Reliability & Availability	<i>Basic data protection</i>	<i>99%</i>
5	Usability & Accessibility	<i>Simple UI</i>	<i>Easy to use</i>
6	<i>Other</i>	<i>Maintainability</i>	<i>Clean code</i>